

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S1 (S,FE) S2 (S,FE) Degree Examination May 2025 (2019 Scheme)

Course Code: MAT102**Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS
(2019-Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

- 1 Find the gradient of $\phi(x, y, z) = xy + yz + zx$ at the point $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$. (3)
- 2 Evaluate the line integral $\int_C (1 + xy^2) dS$ where C is $\vec{r}(t) = t\vec{i} + 2t\vec{j}$, $0 \leq t \leq 1$. (3)
- 3 Determine the sources and sinks of the vector field $\vec{f}(x, y) = xy\vec{i} - 2xy\vec{j} + y^2\vec{k}$. (3)
- 4 Evaluate the surface integral $\iint_{\sigma} (x - y - z) dS$ where σ is the part of the plane $x + y = 1$ that lies in the first octant between $z = 0$ and $z = 2$. (3)
- 5 Solve the initial value problem, $y'' - 9y = 0$, $y(0) = 1$, $y'(0) = 1$. (3)
- 6 Find the Wronskian corresponding to the solution of differential equation $y'' + 25y = 0$ (3)
- 7 Find the Laplace Transform of $\cos 2t \cos t$ (3)
- 8 Find the Laplace Transform of $f(t) = t \delta(t - 1)$ (3)
- 9 Find the Fourier Sine Integral of $f(x) = \begin{cases} a - x & \text{if } 0 < x < a \\ 0 & \text{if } x > a \end{cases}$. (3)
- 10 Find the Fourier Cosine transform of $f(x) = 1$, $0 < x < 1$ (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module-I**

- 11 a) Find the parametric equation of the tangent line to the curve (7)
 $\vec{r}(t) = (3t - 1)\vec{i} + \sqrt{(3t + 4)}\vec{j}$ at $(-1, 2)$
- b) Find the work done by the force $\vec{f}$ on a particle that moves along the curve C if (7)
 $\vec{f} = xy\vec{i} + x^3\vec{j}$ and C is $x = y^2$ from $(0, 0)$ to $(1, 1)$

- 12 a) If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = |\vec{r}|$ prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$ (7)
- b) Show that the vector field $\vec{f}(x, y) = 3x^2e^y\vec{i} + x^3e^y\vec{j}$ is conservative and find ϕ such that $\vec{f} = \nabla\phi$. Hence evaluate $\int_{(0,0)}^{(3,2)} 3x^2e^y dx + x^3e^y dy$. (7)

Module-II

- 13 a) Use the Divergence theorem to find outward flux of $\vec{f}(x, y, z) = (x^2 + y)\vec{i} + xy\vec{j} - (2xz + y)\vec{k}$ where σ is the surface of the tetrahedron bounded by $x + y + z = 1$ and the coordinate planes. (7)
- b) Use Green's theorem to evaluate $\int_C (xy + y^2) dx + x^2 dy$ where C is the boundary of the area common to the curves $y = x^2$ and $y = x$. (7)
- 14 a) Find the flux of the vector field $\vec{f} = x\vec{i} + y\vec{j} + z\vec{k}$ across the portion of the surface $z = 1 - x^2 - y^2$ that lies above the XY-plane oriented by unit normal with positive components. (7)
- b) Use Stokes theorem to evaluate $\oint_C \vec{f} \cdot d\vec{r}$ where $\vec{f}(x, y) = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ and C is the rectangle in the XY-plane bounded by the lines $x = 0, y = 0, x = a, y = b$. (7)

Module-III

- 15 a) Using the method of undetermined coefficients, solve $y'' - 2y' + 5y = x^2$ (7)
- b) By the method of variation of parameter, solve $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ (7)
- 16 a) Solve the initial value problem $x^2y'' + 3xy' + y = 0, y(1) = 0, y'(1) = 1$. (7)
- b) Using the method of undetermined coefficients, solve $y'' - 4y' + 3y = \sin 3x$. (7)

Module-IV

- 17 a) Using Laplace Transform solve $y'' + 9y = 10e^{-t}, y(0) = 0, y'(0) = 0$ (7)
- b) Using convolution theorem, find the inverse Laplace transform of $\frac{a}{s^2(s^2 - a^2)}$. (7)

- 18 a) Express the function $f(t) = \begin{cases} 2, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \\ \sin t, & t > 2\pi \end{cases}$ in terms of unit step function (7)
and hence find its Laplace Transform.
- b) Find the inverse Laplace Transform of $\frac{4s+5}{(s+1)^2(s-2)}$ (7)

Module-V

- 19 a) Using Fourier integral representation show that (7)
- $$\int_0^{\infty} \frac{\cos xw + w \sin xw}{1+w^2} dw = \begin{cases} 0 & \text{if } x < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \\ \pi e^{-x} & \text{if } x > 0 \end{cases}$$
- b) Find the Fourier Transform of $f(x) = \begin{cases} 1-x^2 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1 \end{cases}$ (7)
- 20 a) Represent $f(x) = \begin{cases} e^x, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$ as a Fourier Cosine integral (7)
- b) Find the Fourier Sine Transform of $f(x) = \begin{cases} x, & \text{if } 0 < x < 1 \\ 2-x & \text{if } 1 < x < 2 \\ 0 & \text{if } x \geq 2 \end{cases}$ (7)
